Hierarchical Feature Integration Network for Bi-Temporal Remote Sensing Change Detection Under Complex Environmental Conditions

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***Abstract*—Change detection in high-resolution remote sensing images is affected by various complex factors. Our principal component analysis (PCA) analysis reveals distinct feature clustering in normal environments, contrasting with overlapping, blurred-boundary distributions in complex environments. This blurring arises from seasonal variations, weather conditions, loss of fine-grained textural information, and the inherent difficulty of detecting changed areas. To address these challenges, we propose a hierarchical feature integration network inspired by the human visual system (HVS), integrating model-driven and data-driven approaches by mimicking the HVS's selective and nonselective pathways. Specifically, curvature-sensitive and fractional-order modules, simulating the nonselective pathway's capacity for rapid geometric and textural processing, enhance edges via curvature flow equations and detect fine-grained textural variations using fractional-order parameters. Then, we design a cognitive attention-based data-driven module to simulate the selective pathway of HVS, adjusting feature weights for complex environment adaptation. Finally, a visual area V2-inspired knowledge detection head (KDH) integrates the fast, low-level features from the nonselective pathway and the slow, high-level features from the selective pathway, using an integrated loss function that balances interpretability and learning capacity, achieving $F1$-scores of 93.33%, 94.17%, 90.15%, and 90.61% on LEVIR, DSIFN, CDD, and SYS datasets, respectively, outperforming nine competing methods with improvements up to +11.69% in F1, while maintaining a lightweight architecture of 11.45-M parameters and 26.72-G FLOPs, providing a robust solution for high-resolution remote sensing image change detection (HRRS-CD) in complex environments. Codes are available at our https://github.com/Zzzz12349/HVS-CHANGE**

***Index Terms*—Change detection, complex environments, human visual system (HVS), remote sensing image.**

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I. Introduction

HIGH-RESOLUTION remote sensing image change detection (HRRS-CD) contends with complex environments characterized by the simultaneous occurrence of seasonal variations, weather conditions, and diverse human activities [1], [2], [3]. Unlike ideal environments dominated by a single variable, real-world complex environments involve the interplay of these factors. For instance, a scene may exhibit vegetation senescence due to seasonal changes, illumination shifts caused by weather, and structural updates from human construction [4], [5], [6]. However, when these factors coexist, they introduce severe feature confusion that cannot be resolved by simple augmentation. They lead to varying atmospheric scattering and spectral shifts across bi-temporal images, while the spatial scale similarity between changed and pseudo-changed areas complicates feature discrimination. This ultimately results in blurred boundaries, loss of fine-grained details, and misclassification of unknown areas [7], [8].

To better understand the specific challenges posed by these complex environments mentioned above, we analyze typical cases in Fig. 1. The top row highlights blurred boundaries of changed areas, where the detected regions exhibit unclear boundaries (indicated by red boxes), leading to pseudo-changes (indicated by blue boxes). In the middle row, the focus is on fine-grained changed areas, which are intricate and detailed. Although the baseline method identifies numerous changed areas, it fails to detect certain fine-grained areas (highlighted by red boxes), suggesting that relying solely on learned features is insufficient to capture intricate geometric structures [9]. The bottom row demonstrates that distinguishing true changes from pseudo-changes in complex environments requires high-level semantic discrimination. For example, while shadows over the swimming pool causing pseudo-changes (highlighted by blue boxes) can theoretically be addressed through radiometric normalization, true small-scale changed areas such as chairs and sunshades (red boxes) are missed due to their limited spatial extent falling below conventional detection thresholds. Crucially, the pseudo-change regions and true small-scale changes exhibit similar spatial dimensions and scales. When applying data augmentation or uniform multiscale feature enhancement to improve detection of small true changes, same-scale pseudo-changes are equally amplified, making discrimination in feature space extremely difficult [10].

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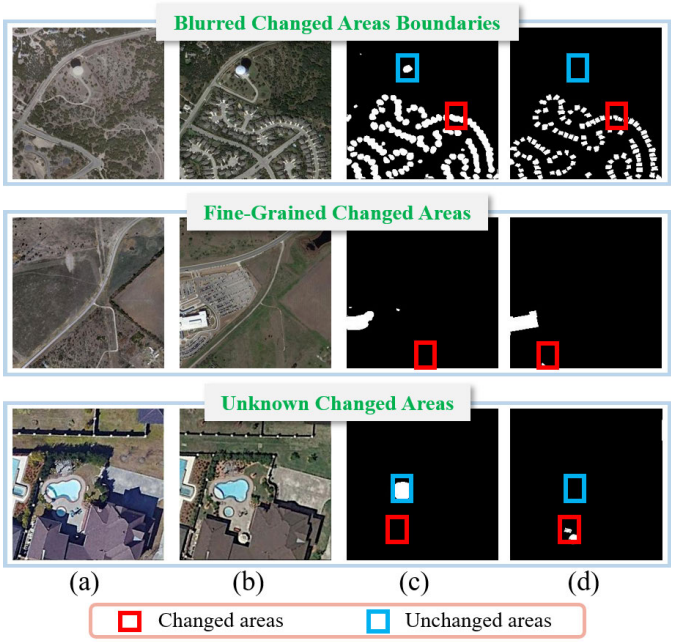


Fig. 1. CD results of HRRS: analysis of blurred, fine-grained, and unknown change areas. (a) HR images at time T1. (b) HR images at time T2. (c) CD results generated by the deep learning method. (d) Ground truth labels.

Existing CD methods are generally categorized into model-driven and data-driven approaches. Model-driven methods rely on mathematical definitions and physical priors, perform well preserving geometric edges, but lack adaptability in semantic complex environments [11], [12]. Data-driven methods, particularly deep learning techniques, demonstrate powerful detection and semantic representation capabilities; however, they may experience boundary blurring and performance degradation when geometric priors are neglected [13], [14]. Despite attempts to combine these approaches, a critical methodological limitation remains: existing strategies typically employ sequential concatenation rather than systematic integration. As highlighted in our analysis, when pseudo-changes and true changes occupy similar spatial frequencies, sequential concatenation fails to selectively filter noise, propagating errors through the network. Furthermore, while biologically motivated methods exist, they adopt partial visual mechanisms without systematically mapping to the hierarchical pathways of the human visual system (HVS).

These limitations collectively contribute to the performance challenges observed in complex environments, where feature distribution overlap complicates conventional detection approaches. The systematic nature of these challenges suggests that solutions require integrated approaches that can simultaneously handle interpretability, scale-discrimination, and biological correspondence. In this context, the HVS offers a proven paradigm for addressing such complex visual processing tasks [15], [16], [17], [18], [19].

Within HVS, the nonselective pathway in the primary visual cortex (V1) is responsible for detecting fundamental geometric features, such as edges, textures, and geometric structures, reducing the boundary blur problem; the selective pathway in the higher level visual cortex (V4) is attention-driven, specializes in processing colors, motion-induced pseudo-change areas, and complex semantic features; and the secondary visual cortex (V2) further refines these changed areas and integrates them with global information through bidirectional feedback mechanisms, serving as a bridge between low-level and high-level processing [20]. The collaborative operations of V1, V2, and V4 enable the HVS to balance local details with global semantics, achieving precise perception of changed regions [21].

Inspired by the selective and nonselective pathway mechanisms of HVS [22], we propose a novel method for HRRS-CD that systematically integrates model-driven and data-driven approaches. Unlike previous methods that loosely reference biological vision [23], [24], our design explicitly maps distinct regions of the visual cortex to corresponding computational components. Specifically, simulating the nonselective pathway (V1), we design a model-driven branch containing curvature-sensitive and fractional-order modules. A curvature-sensitive module (CSM) detects geometric features and refines blurred boundaries; a fractional-order feature module (FODM) processes multiscale texture features. Simulating the selective pathway (V4), we design a cognitive attention-based data-driven module. This module adaptively adjusts feature weights to handle complex scale confusion and semantic variations. Finally, a knowledge detection head (KDH) simulating the bridging function of the V2. The main contributions of this article are listed.

1) We reveal substantial differences in feature distributions between remote sensing images of normal and complex environments through principal component analysis (PCA). This finding provides a theoretical basis for designing change detection models tailored to complex environments.
2) We present the first HRRS-CD method systematically mimicking the hierarchical processing of visual cortex areas V1, V2, and V4. Our approach leverages the V1 nonselective pathway for boundary and texture detection, complemented by the V4 selective pathway for capturing fine-grained details such as color and motion in unchanged regions.
3) Inspired by the integrative function of visual area V2, we design the KDH. Unlike conventional sequential concatenation, KDH achieves systematic integration by adaptively weighting features from selective and nonselective pathways. This approach balances model interpretability and learning capacity while effectively addressing the unique challenges of HRRS-CD.
4) Extensive experiments on four challenging datasets, encompassing diverse complex environmental factors, validate our proposed method. Our approach demonstrates notable improvements over existing CD methods in these complex environments.

The rest of this article is organized as follows. The related works are reviewed in Section II. Section III describes the proposed method. Sections IV and V introduce the datasets used and implementation details. Section VI analyzes the experimental results. Section VII gives the final conclusions.

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II. Related Work

We review the current state of HRRS-CD research, categorizing existing approaches into non-HVS-based [25], [26] and HVS-based methods [27], [28]. We also briefly describe the selective and nonselective pathway mechanisms of the HVS [29], [30], which serve as the foundation for our proposed method.

A. Non-HVS-Based HRRS-CD

HRRS-CD in complex environments is influenced by multiple disturbance factors, including seasonal variations, weather conditions, and human activities. Unlike environments dominated by a single variation source, complex environments are characterized by the coexistence and coupling of these factors, which introduce pseudo-changes and obscure true land-cover changes. Seasonal vegetation dynamics cause appearance changes across time, while weather changes cause radiometric inconsistencies, and human activities further blur change boundaries and increase spatial ambiguity. The simultaneous presence of these disturbances poses a challenge for accurately distinguishing true changes from pseudo-changes in HRRS-CD.

Consequently, existing HRRS-CD approaches can be categorized into three main paradigms, each with distinct limitations. Model-driven methods rely on manually constructed feature spaces [25], [26], making their performance heavily dependent on parameter tuning and limiting their robustness across varying illumination and surface conditions [31], [32], [33]. While they offer physical interpretability, they lack semantic understanding, resulting in blurred changed region boundaries. Data-driven methods, including convolutional neural networks [34], [35], generative adversarial networks [36], [37], and transformers [38], [39], excel at detecting global semantic information [40], [41]. However, their black-box nature limits physical interpretability of network features [42], [43], and they tend to lose spatial information during feature detection, leading to missed fine-grained changed areas. Vision-inspired methods attempt to leverage certain mechanisms of HVS, but remain constrained in complex environments, struggling to simultaneously handle global semantic and local detail information [44].

Thus, existing methods fail to integrate the interpretability of model-driven approaches, the learning capabilities of data-driven methods, and the strengths of vision-inspired techniques. This inability to effectively combine these strengths results in suboptimal performance for CD in complex environments. A unified approach that addresses these limitations is essential for advancing the accuracy of HRRS-CD frameworks.

B. HVS-Based HRRS-CD and Beyond

HVS-inspired methods offer a promising direction for HRRS-CD. Relevant works exploring this direction include studies on saliency-guided CD (e.g., [45]), attention mechanisms inspired by visual attention (e.g., [46]), and hierarchical feature extraction networks (e.g., [47]). Building upon the concept of leveraging the HVS for HRRS-CD, several key aspects warrant further exploration. Primarily, the inherent parallel processing within the HVS, with distinct pathways for different visual attributes (e.g., form, color, motion-induced pseudo-change), can inform the design of multistream frameworks for CD [48], [49]. These streams could focus on specific change characteristics, such as spectral variations, textural differences, and geometric discrepancies, before integrating their outputs for a comprehensive change image [50], [51]. Furthermore, the HVS's ability to suppress irrelevant information and focus on salient regions can be replicated through attention mechanisms [52], [53]. These mechanisms can guide the network to prioritize areas exhibiting changed regions while downplaying the influence of unchanged regions.

This selective attention improves the accuracy and efficiency of CD, especially in large and complex scenes. In addition, incorporating feedback connections, mimicking the top-down processing in the HVS, can further enhance the performance [27]. Feedback from higher level layers refines the feature representations at lower levels, leading to more accurate and context-aware change detection. Finally, the concept of perceptual loss functions, inspired by the HVS's perception of change, can be explored. These loss functions would prioritize perceptual changes rather than pixel-level differences, leading to meaningful change maps. While challenges remain in translating the complexities of the HVS into computational models, ongoing research continues to unlock the potential of this biologically inspired approach for advancing HRRS-CD. This includes exploring more sophisticated neural network architectures, incorporating domain-specific knowledge, and developing robust evaluation metrics that align with human perception of change. For further reading, research on biologically inspired computer vision [29] and the application of neuroscience principles to remote sensing [30] provide valuable insights.

III. Methodology

We first design the structure and motivation of the proposed method. Next, we provide a detailed explanation of the CSM and FODM, which are based on a model-driven approach. In addition, the CAM follows a data-driven approach. Finally, we elaborate on the KDH and the loss function.

A. Overview of Proposed Method

Fig. 2 demonstrates this difference between normal and complex environmental conditions. While normal conditions (left) show clean, well-defined change detection results despite seasonal variations, complex conditions (right) exhibit fragmented and noisy detection patterns. Specifically, the second column shows heavy snow coverage creating homogeneous bright regions that mask underlying features, the third column illustrates ice formation producing specular reflections that dramatically alter surface changes, and fine-grained pseudo-changes are scattered throughout the left upper regions of complex examples. These extreme conditions generate numerous artifacts that challenge traditional detection algorithms. However, when seasonal variations, weather changes, and human activities occur simultaneously, the detection results

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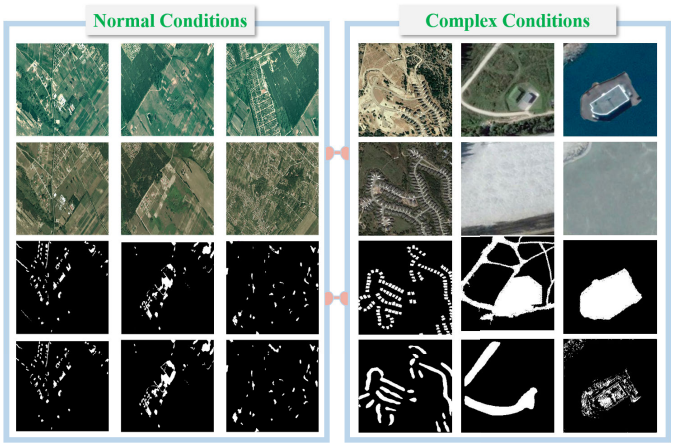


Fig. 2. CD results of HRRS under normal and complex conditions. Normal conditions refer to environments with predictable seasonal variations under stable atmospheric conditions, while complex conditions simultaneously include heavy snow coverage (second column), ice formation (third column), and fine-grained pseudo-changes that create visual feature confusion beyond normal seasonal patterns. (First row: HR images at time T1, second row: HR images at time T2, third row: ground truth labels, and fourth row: CD results generated by the method.)

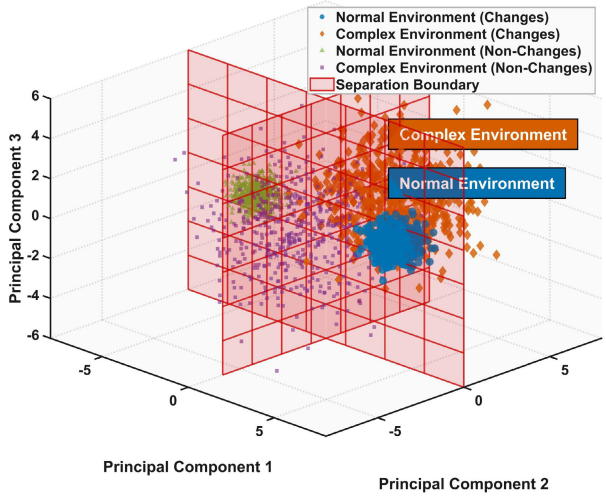


Fig. 3. Visualization of the clusters of the attention heat map (e.g., several examples are shown in Fig. 2) using PCA.

become less accurate. Our experiments confirm that detection accuracy decreases in complex environments, with true changes and pseudo-change areas frequently causing misclassifications. PCA visualization further reveals distinct clustering patterns in Fig. 3: in normal environments, change features (blue circles) and nonchange features (green triangles) form compact, well-separated clusters; while in complex environments, change features (orange diamonds) and nonchange features (purple squares) reveal dispersed distributions with overlapping boundaries, indicating reduced discriminability. This feature distribution difference directly leads to decreased detection accuracy in complex environments, shown in Fig. 2.

While CD in complex environments, as illustrated in Fig. 2, poses challenges for existing methods, the HVS performs this task with relative ease. The HVS uses contextual background perception to identify changes, rapidly analyzing primary hierarchical structural information via the nonselective pathway to establish overall scene structure and category. This information, combined with the selective pathway mechanism, guides more detailed scene feature analysis, facilitating accurate CD.

Inspired by the hierarchical processing of the HVS [54], [55], we propose a novel CD framework, illustrated in Fig. 4, that mimics visual cortex areas V1, V2, and V4 to efficiently integrate global semantic information with local details. This framework addresses the bi-temporal image analysis challenge by focusing on edge and texture discrepancies (V1), semantic understanding (V4), and feature integration (V2).

Our four-module framework comprises a V1-inspired CSM for edge enhancement, a V1-simulated FODM for multiscale feature processing, a V4-inspired CAM for semantic representation enhancement, and a V2-inspired KDH for feature fusion. This integrated approach effectively addresses HRRS-CD by concurrently processing geometric and textural information, adaptively focusing on changed areas, and suppressing pseudo-change areas.

B. Nonselective Pathway Mechanisms Inspired Module

Inspired by the nonselective pathway of the V1, we propose the CSM. In the HVS, the V1 nonselective pathway rapidly processes fundamental features like edges and boundaries without detailed analysis, enabling quick detection of visually changed areas. Similarly, our CSM detects fundamental features and suppresses background noise while simultaneously reducing unchanged areas from the detection results, mimicking this biological mechanism's efficiency and selectivity.

The CSM is based on a geometric partial differential equation known as the mean curvature flow. For a pre- or after-image $I(x, y, t)$, the evolution equation is expressed as

$$F_{CSM}(I_k) = I_k + \sum_{n=0}^{N-1} \Delta t \cdot \kappa\left(I_k^n\right) \cdot |\nabla I_k^n|_\epsilon \tag{1}$$

where I_k^n represents the image at the nth iteration of the discretized mean curvature flow applied to I_k, Δt is the time step, and $|\nabla I_k^n|_\epsilon$ is a regularized gradient magnitude with a small constant ϵ to avoid division by zero. This discretization directly implements the continuous formulation in (1).

The CSM exhibits several properties that make it particularly effective for HRRS-CD. The curvature-based CSM enhances edges while suppressing noise, crucial for handling changes in HRRS. Its adaptive processing preserves sharp edges while smoothing noisy areas, making it suitable for diverse changed areas. Inherently scale and rotation-invariant, the CSM maintains crucial structural details lost with linear filtering, effectively capturing changed areas like building boundaries, road networks, and coastlines. Fig. 5 illustrates the effect of the CSM in HRRS.

Despite its effectiveness, the CSM has limitations when dealing with complex textures found in agricultural areas and transportation networks, as it primarily focuses on geometric information with limited sensitivity to directional and textural patterns. As shown in Fig. 5(a), when there are both farmland and buildings, the CSM struggles to accurately detect changes

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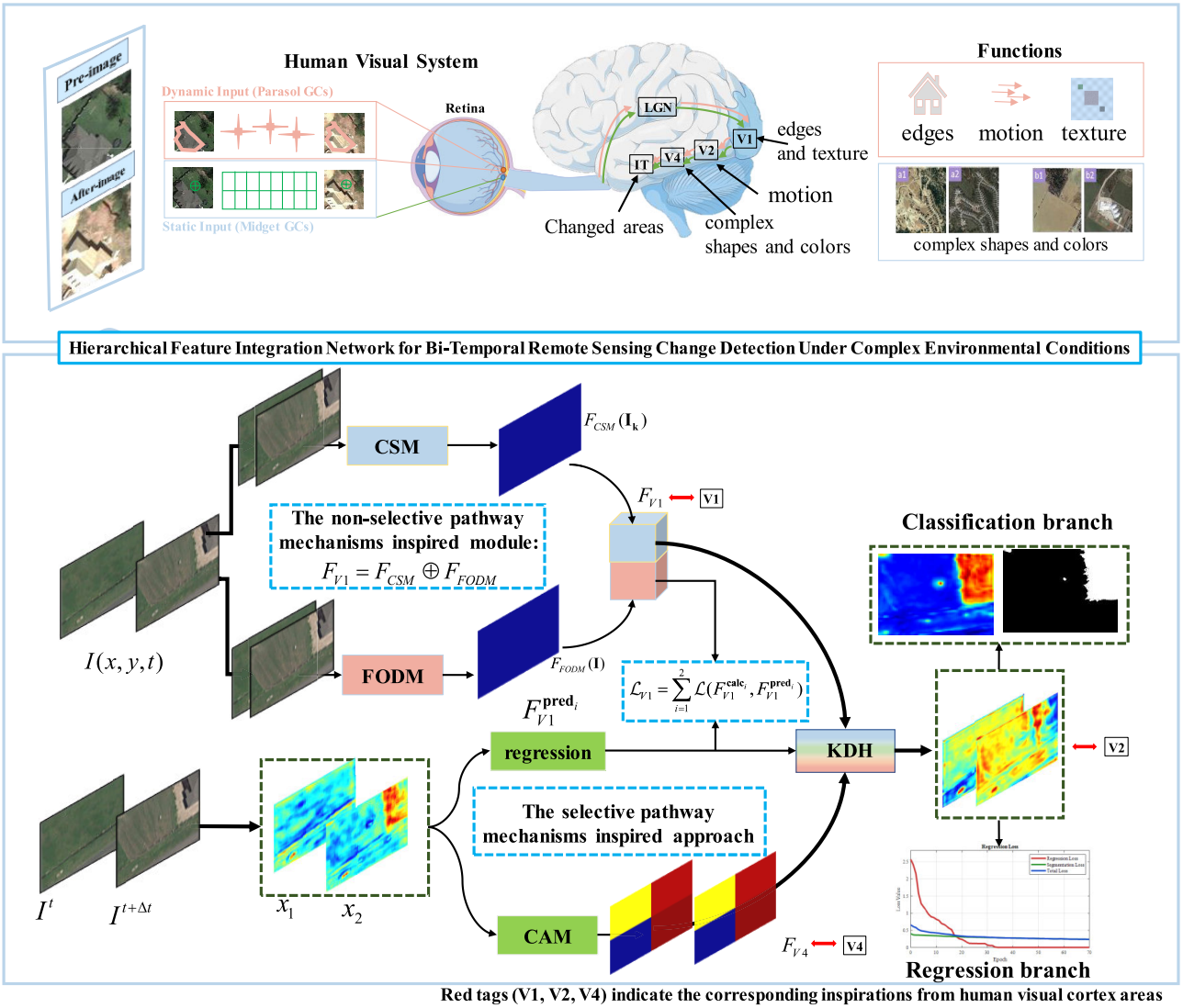


Fig. 4. Overall framework of biological vision-inspired model for complex remote sensing image change detection.

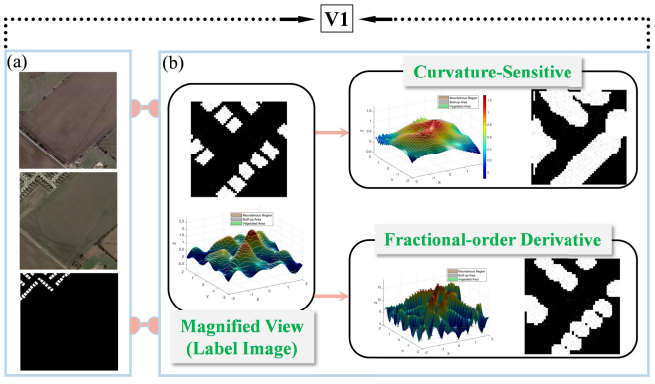


Fig. 5. Example of the nonselective pathway mechanisms inspired module. (a) Original input images. (b) Zoomed-in views of (a) after CSM and FODM enhancement.

in the buildings. As shown in Fig. 5(b), while the main boundaries of the buildings are detected, through the enhanced processing of the CSM image, complex texture areas such as gardens (unchanged areas) in the middle of the buildings, which are similar to the buildings themselves, are identified as changed areas. Hence, we further design additional complementary modules (i.e., FODM) in our framework that specifically handle textural features. FODM extends classical integer-order calculus by allowing derivatives of arbitrary real order. For an input image $I(x, y, t)$, the fractional-order derivative of order $\alpha \in \mathbb{R}^+$ can be expressed using the Grünwald–Letnikov definition

$$D^{\alpha}[I(x)] = \sum_{k=0}^{M} (-1)^k \cdot C(\alpha, k) \cdot I(x-k) \tag{2}$$

where $C(\alpha, k)$ represents the generalized binomial coefficients, and M is the truncation parameter. For 2-D images, we apply separable fractional-order derivatives along both directions and compute the gradient magnitude as

$$|\nabla^{\alpha} I| = \sqrt{\left(D_x^{\alpha}[I]\right)^2 + \left(D_y^{\alpha}[I]\right)^2}. \tag{3}$$

The output of this module is defined as

$$F_{FODM}(I) = \{|\nabla^{\alpha_i} I| \mid \alpha_i \in \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}\} \tag{4}$$

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where α_i represents different fractional orders, each of which continuously controls the frequency sensitivity of the derivative operator, thereby physically decomposing the image's spatial information into four distinct frequency-sensitive channels. Specifically, a fractional-order derivative of order α interpolates continuously between smoothing ($\alpha \approx 0$) and differentiation ($\alpha \geq 1$), so that each $|\nabla^{\alpha_i} I|$ in the set physically represents the image's spatial variation measured at a distinct frequency band. In our experiments, $\alpha_1 = 0.3$ acts closer to a smoothing filter than a differential operator, suppressing high-frequency noise and responding preferentially to large-scale, slowly varying spatial structures (e.g., broad land-cover transitions such as field-to-urban boundaries). $\alpha_2 = 0.7$ balances smoothing and differentiation, making the operator sensitive to medium-scale spatial transitions, such as gradual texture boundaries between vegetation types or semi-urban textures, capturing patterns invisible to both pure smoothing and sharp differentiation. $\alpha_3 = 1.2$, slightly above the standard gradient order ($\alpha = 1$), amplifies crisp edges and structural boundaries while retaining slightly more contextual information than a pure first-order derivative, physically analogous to an edge-sharpening operation with mild surrounding context and suitable for detecting building outlines and road segments. $\alpha_4 = 1.8$, approaching a second-order derivative, responds strongly to fine-grained, high-frequency texture microstructures such as surface roughness in agricultural fields, rooftop patterns, or pavement textures, capturing the rapid local intensity oscillations that distinguish visually similar but semantically different land covers. These values were selected through systematic grid search validation on four dataset training sets to optimize multiscale texture detection performance. Therefore, (4) does not simply describe four independent gradient computations. Instead, it decomposes the image's spatial information into four frequency-sensitive channels, each responding to different orientations and spatial frequencies. This multichannel design is motivated by the scale ambiguity challenge in HRRS-CD. In HRRS images, pseudo-changes and true changes occur at similar spatial scales, making single-order operators insufficient for effective discrimination. By using four gradient directions, the FODM can capture multiscale spatial variations simultaneously. The final feature representation combines both CSM and FODM outputs

$$F_{V1} = F_{CSM} \oplus F_{FODM} \tag{5}$$

where $\oplus$ denotes a channel-wise feature combination operation, inspired by the parallel processing pathways of the primary visual cortex. This operation integrates distinct feature streams to achieve comprehensive visual perception.

1) Texture Sensitivity: Unlike integer-order derivatives that primarily respond to sharp transitions, fractional-order derivatives capture intermediate textural patterns prevalent in complex environments, such as agricultural fields, forests, and urban fabrics.

2) Memory Effect: The nonlocal nature of fractional derivatives inherently incorporates spatial context, as evidenced by the summation over neighborhood points. This property enables the FODM to better distinguish between true changes and pseudo-changes caused by illumination variations or seasonal differences.

3) Continuous Transition Between Smoothing and Differentiation: Fractional calculus provides a continuous spectrum between smoothing operations ($\alpha \approx 0$) and differentiation ($\alpha \geq 1$), allowing for more nuanced feature detection compared to conventional techniques that typically combine distinct smoothing and differentiation steps. Fig. 5 shows the effectiveness of the FODM across different fractional orders.

In short, integrating FODM and CSM creates a comprehensive framework for detecting changed areas, capturing both geometric and textural information. While CSM enhances structural boundaries through curvature-based processing, FODM works with multiscale texture features and detects patterns in the texture.

C. Selective Pathway Mechanisms Inspired Approach

Under the model-driven approach, although edges and texture features can be detected through the CSM and FODM modules, the detection is limited by color and shape pseudo-change factors. To address these problems in complex environments, we incorporate selective processing mechanisms inspired by the HVS's V4.

Specifically, a CAM is designed that integrates dynamic feature extraction and adaptive weight modulation to enhance the model's focus on changed regions while reducing pseudo-changes. The process begins with the extraction of high-level features from the input images through a deep neural network, denoted as x_1 and x_2

$$x_1 = F_{\text{deep}}\left(I^t\right), \quad x_2 = F_{\text{deep}}\left(I^{t+\Delta t}\right) \tag{6}$$

where F_{deep} represents the deep feature extractor, which captures rich semantic information from the input images at time t and $t + \Delta t$, respectively. In detail, our deep feature extractor is based on a ResNet architecture that processes input images through initial convolution, batch normalization, ReLU activation, and max pooling operations. The features are then sequentially processed through ResNet's first and second layers, producing feature maps at 1/4 and 1/8 resolution scales. In our proposed approach, the model can optionally utilize ResNet's third and fourth layers for deeper feature extraction. Finally, a $2\times$ upsampling is applied before the prediction convolution layer generates the output features, denoted as x_1 and x_2. These extracted features serve as the input to the subsequent cognitive modulation components.

The global cognitive modulation component captures the overall spatial distribution characteristics of changed regions. Specifically, it computes global weights w_{global} as

$$w_{\text{global}} = \sigma\left(W_2\left(\sigma\left(W_1 x_i\right)\right)\right) \quad \forall i \in \{1, 2\} \tag{7}$$

where W_1 and W_2 are learnable weight matrices, and σ represents the activation function. These global weights highlight important spatial regions in the feature map.

In addition to global modulation, we design the hierarchical cognitive modulation to enhance the model's focus on changed

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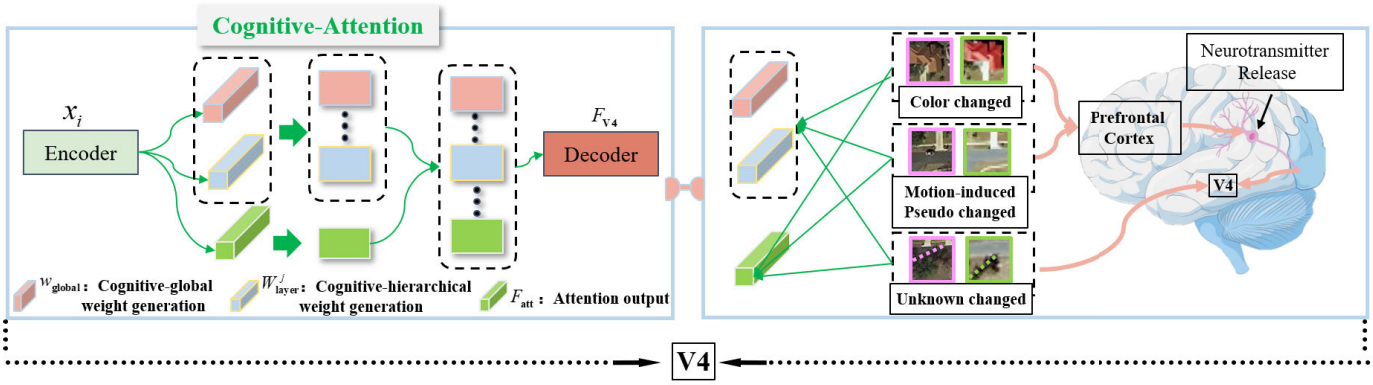


Fig. 6. CAM design inspired by selective pathway mechanisms.

regions through layer-specific weights. For the jth layer, the weights w^j_{layer} are computed as

$$w^j_{\text{layer}} = \sigma\left(W^j_{\text{layer}} x^j_i\right) \tag{8}$$

where W^j_{layer} is the weight generation matrix specific to the jth layer, and x^j_i represents the feature representation from the jth layer. This hierarchical approach ensures focus on fine-grained changed regions.

The final feature output integrates both global and hierarchical cognitive weights. The integration process combines attention-enhanced features F_{att} with the original features x_i as follows:

$$\begin{aligned} F_{\text{V4}} = {} & w_{\text{global}} \cdot w^j_{\text{layer}} \cdot F_{\text{att}} \\ & + \left(1 - w_{\text{global}} \cdot w^j_{\text{layer}}\right) \cdot x_i \end{aligned} \tag{9}$$

where F_{att} represents the output of the attention mechanism. The attention-enhanced features F_{att} are generated through a multihead self-attention mechanism that operates on the deep feature representations. This mechanism enables the model to dynamically focus on relevant spatial locations and feature channels that are most indicative of changes between the bitemporal images. The attention module computes attention weights across different heads and combines them to produce enhanced feature representations that highlight important change-related information while suppressing irrelevant background features. Fig. 6 illustrates the effectiveness of the CAM in detecting changed areas in complex environments.

During experimentation, we observed that directly combining features from the model-driven components (CSM and FODM) and the data-driven component (CAM) led to increased computational complexity and a semantic gap between low-level and high-level features. To mitigate this, we design a KDH that bridges this gap, analogous to the function of V2, while simultaneously optimizing computational efficiency.

The KDH operates in two distinct phases with unified network parameters, as shown in Fig. 7. During training, we calculate V1 visual features directly from the input images using the CSM and FODM modules as defined in (5)

$$F^{\text{calc}}_{V1} = F_{V1}. \tag{10}$$

These calculated features are compared with predicted V1 features, which are generated from deep feature representations using the same regression branch

$$F^{\text{pred}_i}_{V1} = \mathcal{R}\left(x_i\right). \tag{11}$$

The regression loss $\mathcal{L}_{V1}$ guides the model to learn the mapping between deep features and V1 features during training

$$\mathcal{L}_{V1} = \sum_{i=1}^{2} \mathcal{L}\left(F^{\text{calc}_i}_{V1}, F^{\text{pred}_i}_{V1}\right). \tag{12}$$

For both training and testing phases, the feature processing follows the same pipeline with unified parameters. For $i \in \{1, 2\}$

$$x^{\text{dec}}_i = \mathcal{D}\left(\mathcal{F}_{\text{fusion}}\left(x_i \oplus F^{\text{calc}_i}_{V1}\right), F^i_{V4}\right). \tag{13}$$

The change detection map is consistently generated using

$$C_{changed} = \mathcal{C}\left(\text{Upsample}\left(\left|x^{\text{dec}}_1 - x^{\text{dec}}_2\right|\right)\right). \tag{14}$$

The key difference lies only in loss computation: during training, the regression loss $\mathcal{L}_{V1}$ is calculated and backpropagated, while during testing, this loss computation is omitted and the model directly outputs the classification result $C_{changed}$ using the same network parameters learned during training.

D. Loss Function

To overcome limitations of traditional segmentation-based loss functions in HRRS-CD (focusing solely on pixel-level accuracy while ignoring spatial context, struggling with multiscale features, and sensitivity to class imbalance), we propose a hierarchically integrated loss function inspired by the HVS.

1) Base Segmentation Loss: The base segmentation loss enhances adaptability to varying region sizes and focuses attention on difficult-to-detect areas (e.g., unknown changed areas)

$$\mathcal{L}_{base} = \lambda_{ce} \cdot \mathcal{L}^{focal}_{CE} + \lambda_{ms} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}^i_{scale} \tag{15}$$

where $\lambda_{ce} = 0.6$ and $\lambda_{ms} = 0.4$ are weighting coefficients that balance focal loss attention with multiscale consistency, and N represents the number of scales. The focal loss $\mathcal{L}^{focal}_{CE}$ automatically reduces the weight of easily classified samples,

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allowing the model to focus on difficult-to-classify regions. The multiscale consistency term $\mathcal{L}^{i}_{scale}$ is applied to multiscale predictions rather than multiscale features. At each scale i, the full-resolution logit map is downsampled via bilinear interpolation to produce the scale-specific prediction $\hat{P}_i$, and the corresponding ground-truth label map is downsampled using nearest-neighbor interpolation to keep the discrete class labels intact. A focal loss is then computed independently at each scale as $\mathcal{L}^{i}_{scale} = \mathcal{L}^{focal}_{CE}(\hat{P}_i, Y_i)$, where Y_i is the downsampled ground-truth label map at scale i. The $1/N$ averaging factor ensures that the total multiscale contribution does not grow as the number of scales N increases, so the loss magnitude remains independent of the choice of N. This multiscale design ensures that features at different hierarchical levels maintain coherent representations, which is important for preserving fine-grained changed regions.

2) Cognitive Regularization Strategy: Drawing from biological visual processing principles, we implement a three-component regularization strategy

$$\begin{aligned}\mathcal{L}_{regularization} = {} & \alpha \cdot \mathrm{D}_{KL}\left(\mathrm{Softmax}\left(F_{V1}/T\right), \mathrm{Softmax}\left(F_{V4}/T\right)\right) \\ & + \beta \cdot \mathrm{mse}\left(F_{V1}, F_{V4}\right) \\ & + \gamma \cdot \left\| F_{V1} - F^{\mathrm{pred}}_{V1} \right\|_2 . \end{aligned} \tag{16}$$

The first term uses KL divergence to align the output class-probability distributions of F_{V1} and F_{V4} at the output distribution level. A softening temperature $T = 2.0$ is applied to both distributions before computing the divergence, which helps the model preserve the relative class relationships encoded in the higher level features. This output-level alignment mimics the feedback from V4 to V1 areas in the HVS, where semantic information flows from higher processing levels to lower ones in HRRS-CD. The second term uses mse to enforce consistency between F_{V1} and F_{V4} at the intermediate feature representation level. Output-level distribution alignment does not constrain the internal feature representations, because a model can maintain stable output distributions while its intermediate feature maps drift during training. This feature drift can weaken the feature extractor when the model encounters new datasets. The mse term directly addresses this problem by penalizing differences between the feature maps at the two processing levels, which corresponds to the lateral connections between different visual processing streams that support information integration. The third term enforces consistency between the calculated and predicted V1 features, reflecting the processing stability observed across visual cortex levels. Parameters $\alpha = 0.4$, $\beta = 0.3$, and $\gamma = 0.3$ control the relative importance of the output-level distillation, feature-level alignment, and feature consistency terms, respectively. The overall regularization weight $\lambda_{reg} = 0.3$ balances conventional detection requirements with the hierarchical processing constraints.

To validate the contribution of each regularization term, we conduct an ablation study on the LEVIR dataset, as shown in Table I. Removing both the KL and mse terms leads to a clear drop in all metrics, which confirms that the regularization terms are necessary. Using only the KL term improves performance over the baseline, but the gain is smaller than when using only the mse term in terms of mean intersection over union (mIoU), which supports the conclusion that output-level alignment alone is insufficient without feature-level constraints. The combination of both terms achieves the best results across all metrics, demonstrating that they provide complementary benefits.

TABLE I
ABLATION STUDY ON THE REGULARIZATION TERMS IN (16)

Configuration	OA (%)	F1 (%)	mIoU (%)
w/o KL, w/o MSE	94.02	79.06	59.02
KL only	95.98	86.41	75.18
MSE only	94.70	87.47	79.56
KL + MSE	**96.83**	**93.33**	**83.88**

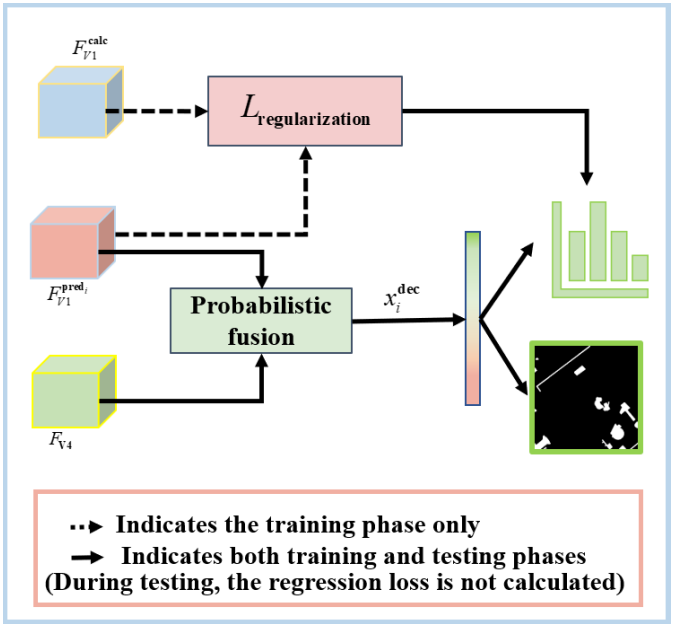


Fig. 7. KDH's role in processing visual information. The figure illustrates how V1 features (basic visual processing) and V4 features (higher level integration) interact through the designed V2-like mechanism.

3) Integrated Optimization Framework: The final optimization objective balances conventional detection requirements with hierarchical processing

$$\mathcal{L}_{total} = \left(1 - \lambda_{reg}\right) \cdot \mathcal{L}_{base} + \lambda_{reg} \cdot \mathcal{L}_{regularization}. \tag{17}$$

IV. DATASET AND EXPERIMENTAL SETTINGS

A. Datasets and Implementation Details

The LEVIR dataset collects 637 high-resolution building change pairs marked at the instance level, focusing on urban CD due to human interference such as construction and demolition. The SYS dataset contains 20 000 pairs of aerial images of size 256×256 taken between 2007 and 2014 in Hong Kong, capturing diverse change types including newly built urban buildings, suburban dilation, groundwork before construction, vegetation changes, road expansion, and sea construction [56]. The DSIFN dataset addresses weather-related challenges like haze and cloud cover, providing pixel-level annotations for changes in both agricultural and urban areas [57]. The CDD

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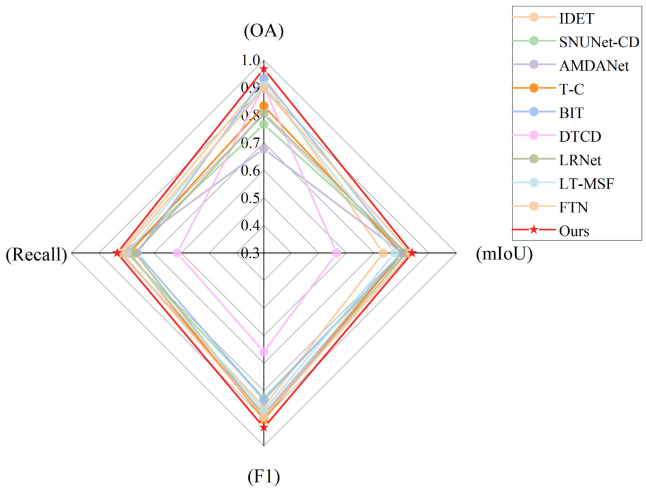


Fig. 8. Radar chart of results on the LEVIR dataset. The red line represents our method, highlighting its effectiveness in distinguishing between unchanged areas and pseudo-changes from human activities.

TABLE II
SENSITIVITY ANALYSIS OF λ_{reg} ON THE LEVIR DATASET

λ_{reg}	OA (%)	F1 (%)	mIoU (%)
0.1	94.74	88.87	78.98
0.2	94.64	89.12	79.21
0.3	**95.52**	**90.69**	**81.07**
0.4	95.23	90.06	80.33
0.5	94.88	89.52	79.68

dataset serves as a public benchmark [58], the dataset captures a variety of changes, including the construction of buildings and roads, while excluding those due to seasonal variations in vegetation and snow cover, as illustrated in Figs. 8–13.

B. Optimization

The model is trained using the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 16 for all datasets. The model is implemented in PyTorch and trained on a single NVIDIA GeForce RTX 5070 Ti with 16 GB of GPU memory. The network is randomly initialized and then trained using the proposed loss function.

All loss weighting hyperparameters were determined through a grid search on the validation split of each dataset. For the balance coefficients λ_{ce} and λ_{ms}, a search was performed over $\lambda_{ce} \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ with the constraint $\lambda_{ms} = 1 - \lambda_{ce}$, yielding the best values $\lambda_{ce} = 0.6$ and $\lambda_{ms} = 0.4$. For the regularization weight λ_{reg}, a search was performed over the set $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. As shown in Table II, $\lambda_{reg} = 0.3$ consistently achieves the best validation performance across all three metrics, and values that are either too small or too large lead to lower performance, which suggests that a balance between the base segmentation loss and the regularization terms is important. For the weights α, β, and γ, a coarse search was performed over $\{0.1, 0.2, 0.3, 0.4, 0.5\}$ with all other parameters fixed at their best-found values, yielding $\alpha = 0.4$, $\beta = 0.3$, and $\gamma = 0.3$. A sequential search strategy was used due to the computational cost of a joint search across all parameters.

TABLE III
COMPARISON OF PERFORMANCE METRICS. "BLURRED" REPRESENTS THE EVALUATION OF BLURRED CHANGED AREA BOUNDARIES (LEVIR DATASET). "FINE-GRAINED" REPRESENTS THE EVALUATION OF FINE-GRAINED CHANGED AREAS (CDD AND SYS DATASETS). "UNKNOWN" REPRESENTS THE EVALUATION OF UNKNOWN CHANGED AREAS (DSIFN DATASET)

Algorithm (Year)	Blurred	Fine-grained	Unknown
IDET (2025) [59]	×	×	×
SNUNet-CD (2022) [58]	×	×	×
AMDANet (2025) [60]	×	×	×
T-C (2022) [61]	✓	×	✓
BIT (2022) [62]	✓	✓	×
DTCD (2020) [63]	×	✓	×
LRNet (2024) [64]	×	×	×
LT-MSF (2025) [65]	×	×	×
FTNet (2022) [66]	×	×	×
Ours (-)	✓	✓	✓

Note: ✓ indicates that the method ranks in the top-2 for both F1 or OA metrics in the corresponding dataset (see Table IV). × indicates the method does not meet this criterion. This binary evaluation focuses on methods that achieve balanced high performance across key metrics.

C. Quantitative Evaluation Metrics

To evaluate the performance of models in HRRS-CD, we selected four key metrics: overall accuracy (OA), mIoU, $F1$-score, which is the harmonic mean of precision and recall for changed areas, and recall for the change class (recall). OA reflects the model's general classification performance, while mIoU evaluates the spatial overlap quality between changed and unchanged regions, emphasizing boundary segmentation precision. F1 combines precision and recall to assess the detection ability for sparsely distributed changed areas, and recall measures the model's capability to capture changed regions, reducing pseudo-change detection [9].

V. EXPERIMENTAL RESULTS

A. Comparison With SOTAs

The experiments presented in Table III demonstrate the performance of our proposed model across multiple CD environments. While existing state-of-the-art (SOTA) methods like IDET, LRNet, and DTCD excel in specific evaluation contexts, our approach uniquely achieves optimal performance in four evaluation scenarios: blurred boundaries between changed and unchanged areas (LEVIR dataset), fine-grained changed areas (CDD and SYS datasets), and unknown changed areas (DSIFN dataset). Our approach uniquely achieves optimal performance in four evaluation scenarios. Specifically, we selected four representative datasets to validate these environments: LEVIR represents blurred boundary challenges from seasonal and lighting variations, CDD and SYS focus on fine-grained urban change detection requirements, and DSIFN evaluates performance under diverse geographical conditions with unknown change patterns.

To further quantify these improvements and understand the performance of the model designed in this article, we conduct

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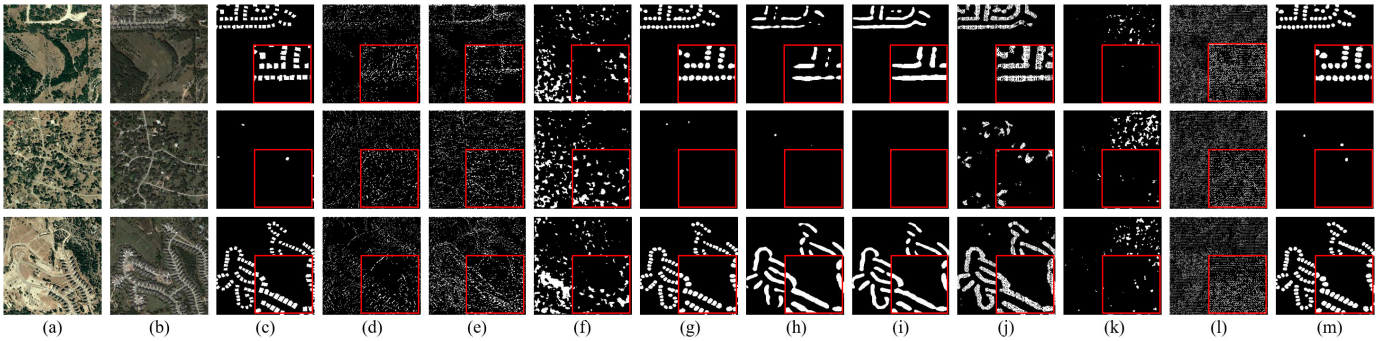


Fig. 9. Qualitative comparison of change detection results on the LEVIR dataset. (a) HR image at time T1. (b) HR image at time T2. (c) Label. (d) IDET. (e) SNUNet-CD. (f) T-C. (g) BIT. (h) DTCD. (i) LRNet. (j) LT-MSF. (k) AMDANet. (l) FTNet. (m) Proposed method.

TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT CD METHODS ON FOUR BENCHMARK DATASETS

Method	LEVIR				DSIFN				CDD				SYS			
	Resolution: 0.5m/pixel				Resolution: 2m/pixel				Resolution: 0.3-1m/pixel				Resolution: 0.5m/pixel			
	OA	mIoU	F1	recall	OA	mIoU	F1	recall	OA	mIoU	F1	recall	OA	mIoU	F1	recall
IDET [59]	76.01	73.34	85.79	82.34	82.73	63.02	76.01	84.02	82.09	81.49	85.62	80.79	87.36	70.66	72.15	89.02
SNUNet-CD [58]	81.68	81.53	82.62	76.70	82.72	62.69	75.69	73.51	82.68	82.53	79.62	78.70	92.33	76.38	76.16	66.70
AMDANet [60]	67.82	80.26	88.46	81.63	81.28	61.07	74.43	72.86	67.82	80.26	82.46	83.63	84.36	50.09	28.05	16.59
T-C [61]	81.36	80.05	90.35	82.89	88.31	81.89	82.48	77.85	80.78	87.50	67.18	82.46	90.46	70.24	67.62	54.17
BIT [62]	93.30	80.20	83.33	76.07	78.53	63.30	80.68	71.84	89.59	73.07	88.49	81.56	93.11	79.52	80.28	76.26
DTCD [63]	90.56	56.44	65.88	61.46	80.24	65.84	45.61	56.48	95.13	68.86	78.66	73.90	93.03	80.03	81.14	81.53
LRNet [64]	80.72	81.75	89.29	76.70	84.64	51.61	61.75	59.50	83.36	67.05	75.35	71.89	89.84	68.81	65.66	52.84
LT-MSF [65]	90.78	77.50	87.18	79.46	82.12	60.37	73.48	70.83	89.30	74.20	73.33	70.07	93.45	80.94	82.06	81.56
FTNet [66]	75.59	83.07	89.56	81.49	67.76	45.48	78.94	69.44	70.56	56.44	65.88	74.46	84.78	52.36	34.10	21.42
Ours	**96.83**	**83.88**	**93.33**	**93.26**	**95.34**	**89.10**	**94.17**	**94.66**	**96.63**	**87.88**	**90.15**	**92.97**	**96.37**	**89.21**	**90.61**	**95.38**
Improvement†	+3.53	+0.81	+2.98	+10.37	+7.03	+7.21	+11.69	+10.64	+1.50	+0.38	+1.66	+9.34	+2.92	+8.27	+8.55	+6.36

†Improvement: percentage points improved by our method compared to the best existing method in each column. All metrics are reported in percentage (%).

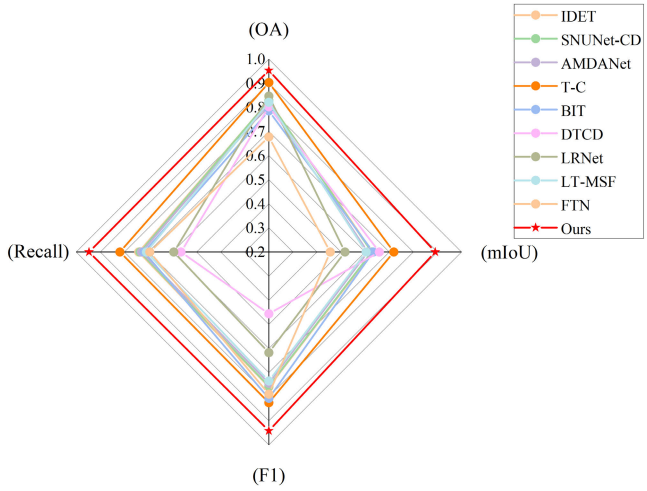


Fig. 10. Radar chart of results on the DSIFN dataset. Our approach effectively identifies unchanged areas despite seasonal and weather variations.

a detailed analysis of the quantitative results across these challenging datasets. Table IV highlights the robustness of our HVS-inspired approach in challenging environments characterized by seasonal changes, weather variations, and human activities. On the LEVIR dataset, which contains pseudo-changes induced by human activities, our method achieves an OA of 96%, outperforming the highest-performing existing method (BIT, OA = 93%) by 3 percentage points. This improvement can be attributed to the CSM and FODM, which improve edge and texture detection while reducing irrelevant pseudo-changes inspired by the V1 region. Furthermore, our method achieves consistent performance across datasets, with minimal performance degradation when transitioning from standard environments (LEVIR) to more complex scenarios (CDD and SYS), indicating robust discriminability for changed areas across diverse challenging conditions.

A critical observation is the balance between OA and recall metrics, as reflected in the $F1$-score. Although methods like SNUNet-CD and LRNet are based on modern deep learning, they still exhibit certain limitations similar to traditional model-driven methods, achieving high OA but lower recall when faced with fine-grained or complex changed regions. Conversely, data-driven methods (e.g., IDET, FTNet, and T-C) prioritize higher recall at the expense of precision. Our method achieves a more balanced performance profile (F1 = 93% on the CDD dataset), attributed to the KDH, which essentially fuses low-level geometric features with high-level semantic features to ensure accurate detection of both large-scale and fine-grained changed areas.

To provide a more intuitive visualization of the comparative performance, we present three radar charts that illustrate our proposed method's performance with nine SOTA approaches. These charts offer a clear representation of the results across

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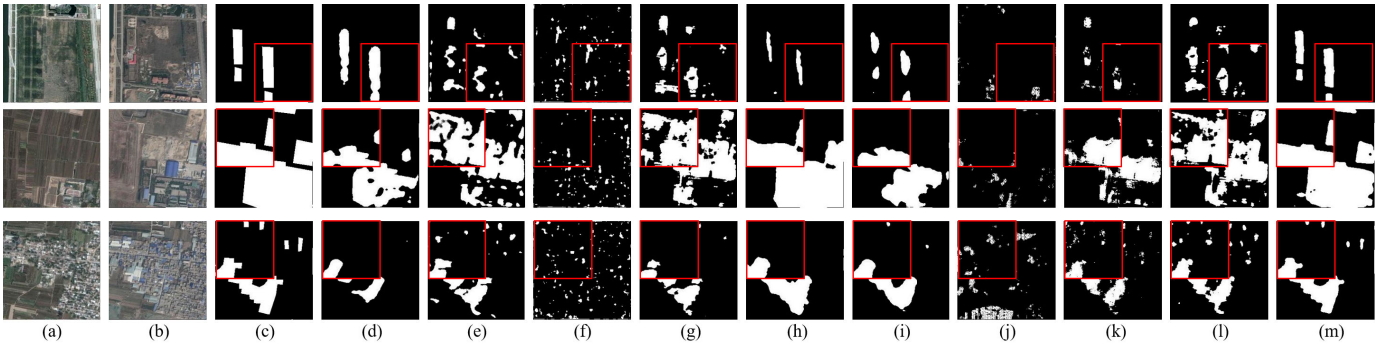


Fig. 11. Qualitative comparison of change detection results on the DSIFN dataset. (a) HR image at time T1. (b) HR image at time T2. (c) Label. (d) IDET. (e) SNUNet-CD. (f) T-C. (g) BIT. (h) DTCD. (i) LRNet. (j) LT-MSF. (k) AMDANet. (l) FTNet. (m) Proposed method.

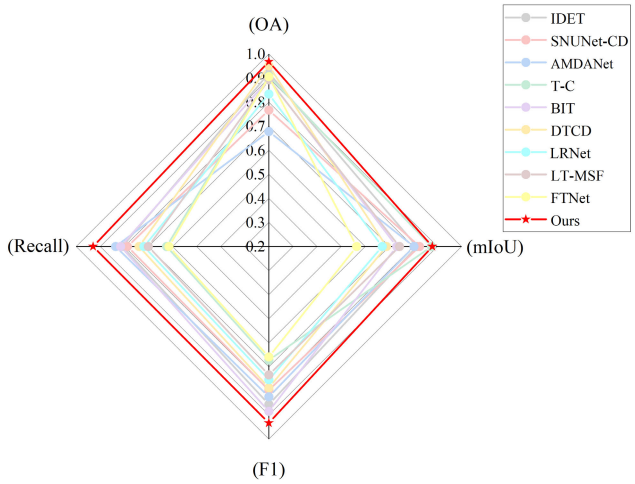


Fig. 12. Radar chart of results on the CDD dataset. The red line represents our method, promising performance in identifying unchanged areas under certain complex environmental conditions.

four datasets and highlight the strengths of our method. Each axis represents a key evaluation metric with all values normalized to a scale of 0–1 for consistent comparison.

Fig. 8 shows results on the LEVIR dataset, characterized by pseudo-changes from human activities. Our method outperforms comparison approaches across all metrics, particularly in OA, and effectively identifies and excludes unchanged regions.

Fig. 10 presents performance on the DSIFN dataset with seasonal and weather variations. Our approach maintains balanced performance across all metrics, highlighting the advantages of our integrated strategy and its adaptability to diverse conditions while effectively removing unchanged regions.

Fig. 12 illustrates results on the CDD dataset, known for complex conditions including varying illumination and diverse land cover types. Despite these challenges, our method achieves high performance across all metrics, demonstrating superior ability in unchanged region exclusion.

While the quantitative results in Table IV clearly reveal the highlights the performance of our HVS-inspired approach across all metrics and datasets, qualitative visual comparison provides further insights into the strengths of each method. Figs. 9, 11, 13, and 14 offer visual comparisons of detection results on the datasets, illustrating how our method performs across varying complex environments.

The LEVIR dataset primarily features pseudo-changes caused by human activities, such as construction or demolition. As shown in Fig. 9, our method essentially distinguishes real changes from pseudo-changes, achieving sharper boundaries and preserving fine-grained changed areas. In the magnified views (where each red box in the small images corresponds to the respective magnified region), it is evident that our approach minimizes false positives (FPs) in regions with minor structural alterations, where other methods produce over-segmentation. This indicates our method's ability to handle human-driven datasets with high precision and reliability.

The DSIFN dataset presents additional complexities with seasonal and weather-induced variations. Fig. 11 illustrates how our method adapts to these diverse environmental conditions while maintaining robust performance. The magnified views in this figure show consistent and accurate detections even in regions with fine-grained changed areas, such as vegetation changes due to seasonal cycles. Unlike other methods that may over-detect or fail to capture fine-grained changed areas, our approach strikes a balance between sensitivity and precision, avoiding unnecessary over-segmentation while capturing meaningful changed areas.

The CDD dataset, shown in Fig. 13, our method achieves accurate detections with well-defined boundaries, as shown in the magnified views. The designed approach shows strong robustness, especially in changed areas with changed light and complex land cover, where other methods fail to separate true changed areas from noise or variations in unchanged areas. This highlights our approach's ability to perform well in real-world environments.

For the SYS dataset, as shown in Fig. 14, our method demonstrates better performance. In the third row, the detected change regions produced by our method exhibit clear and continuous boundaries that are well aligned with the ground truth, while most competing methods suffer from boundary blurring or edge expansion. Moreover, the background areas in our results remain largely clean, with significantly fewer isolated false detections in texture-rich regions. Consequently, our model effectively suppresses pseudo-changes caused by

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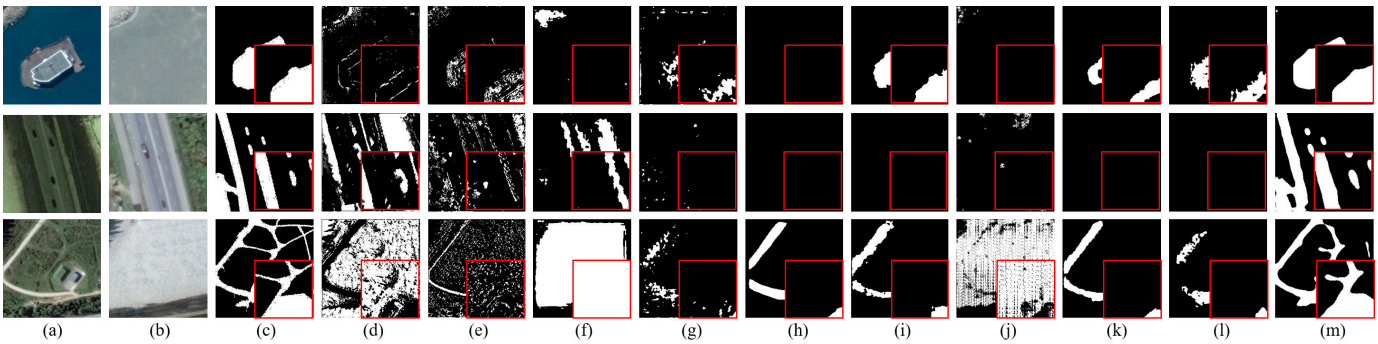


Fig. 13. Qualitative comparison of change detection results on the CDD dataset. (a) HR image at time T1. (b) HR image at time T2. (c) Label. (d) IDET. (e) SNUNet-CD. (f) T-C. (g) BIT. (h) DTCD. (i) LRNet. (j) LT-MSF. (k) AMDANet. (l) FTNet. (m) Proposed method.

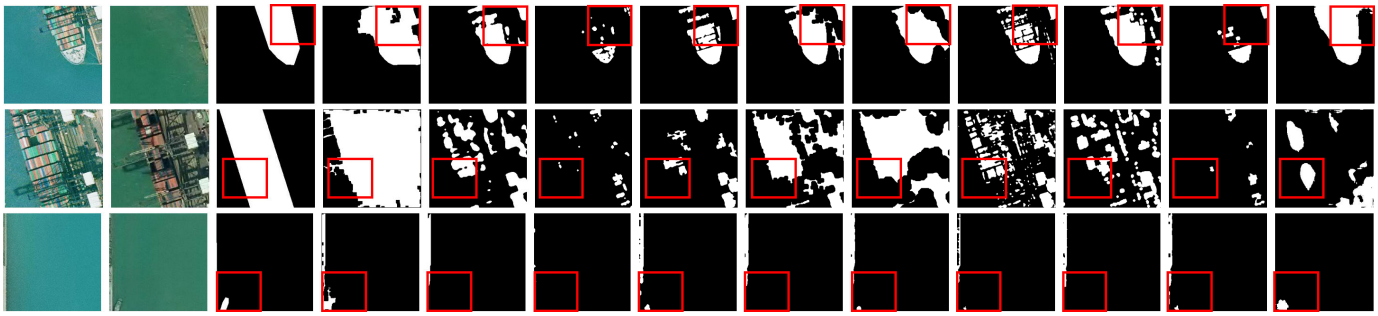


Fig. 14. Qualitative comparison of change detection results on the SYS dataset. (a) HR image at time T1. (b) HR image at time T2. (c) Label. (d) IDET. (e) SNUNet-CD. (f) T-C. (g) BIT. (h) DTCD. (i) LRNet. (j) LT-MSF. (k) AMDANet. (l) FTNet. (m) Proposed method.

complex background textures while preserving the integrity of real change areas.

In summary, these qualitative results provide strong visual evidence of the advantages of our HVS-inspired approach. Across all datasets, our method consistently delivers sharper boundaries, better preservation of fine-grained changed areas, and more robust handling of environmental complexities. These findings align with the quantitative analysis and further validate the effectiveness of our method in addressing HRRS-CD challenges.

Fig. 15 provides a comprehensive comparison of the hierarchically integrated loss function's performance across datasets. The results assess the contributions of the base segmentation loss and regression loss to the overall optimization process. The top row shows the loss trends, the middle row presents the correlation heatmaps, and the bottom row illustrates the total loss decline rates.

1) Loss Trends: The loss trends (top row) reveal a rapid decrease in total loss during the initial training epochs across all datasets. This pattern is primarily driven by the base segmentation loss, which integrates cross-entropy, multiscale, and focal loss components to achieve effective classification and adaptability. For the LEVIR dataset, the regression loss stabilizes quickly at a low value (approximately 0.01), indicating efficient feature alignment across hierarchical levels. On the DSIFN, CDD, and SYS datasets, the segmentation loss continues to decrease gradually, demonstrating the model's ongoing adaptation to challenging multiscale and fine-grained changed areas.

2) Correlation Heatmaps: The correlation heatmaps (middle row) illustrate the relationships between individual loss components and the total loss. On the LEVIR dataset, the regression loss exhibits the strongest correlation with the total loss (correlation coefficient: 91%), highlighting its critical role in aligning low-level (V1) and high-level (V4) changed regions. In contrast, on the DSIFN, CDD, and SYS datasets, the segmentation loss dominates the optimization process (correlation coefficient: 99%), reflecting the importance of multiscale and focal loss contributions for these more complex datasets. These results confirm that the hierarchical integration of segmentation and regression loss components ensures robust convergence across datasets with varying characteristics.

3) Loss Decline Rates: The total loss decline rates (bottom row) show that the highest initial optimization occurs on the CDD dataset (decline rate: 27%), followed by LEVIR (23%) and DSIFN (18%). The rapid optimization in early training stages is primarily driven by the regression loss, which aligns low-level details with high-level semantic changed areas, and the multiscale component of the segmentation loss, which helps the model learn faster by handling both small and large regions.

In conclusion, both qualitative and quantitative results demonstrate that the proposed HVS-inspired approach integrates low-level geometric features with high-level semantic understanding to achieve SOTA performance in HRRS-CD. The hierarchical loss function further enhances optimization. By using the hierarchical processing mechanisms of the HVS, the proposed method achieves an optimal balance between

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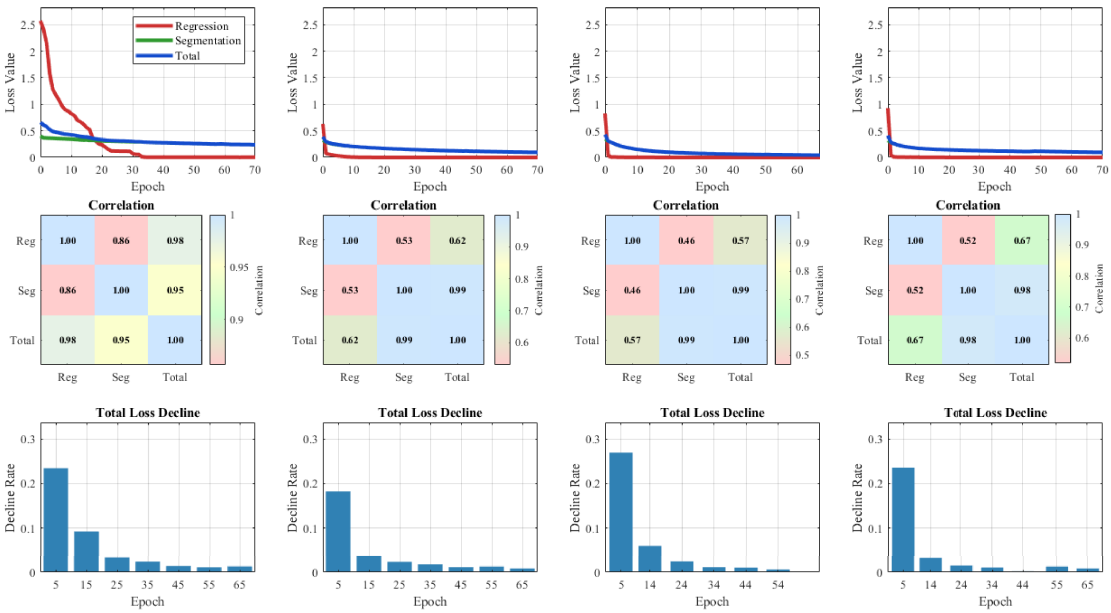


Fig. 15. Comparison of correlation heatmaps, loss trends, and loss decline rates across four datasets. Rows represent different metrics: loss trends (top row), correlation heatmaps (middle row), and loss decline rates (bottom row). Columns represent datasets (LEVIR, DSIFN, CDD, and SYS).

robustness and adaptability, providing a solution for detecting changes across a wide variety of complex environments.

B. Ablation Experiments

We conducted ablation experiments to evaluate the contributions of each module in the proposed system. The CSM baseline module detects geometric features but struggles with directional and texture information. To address these limitations, we design three additional modules: the FODM, the CAM, and the KDH. Each module builds upon the strengths of the previous ones while addressing their limitations, forming a hierarchical architecture inspired by the multipath information processing mechanisms of the HVS.

1) Effect of the CSM: As the foundational module in Fig. 16(a) and (b), CSM focuses on geometric feature detection in HRRS-CD. Employing dynamic curvature perception, nonlinear expansion, and adaptive smoothing, it enhances edge features and suppresses unchanged areas. These capabilities handle geometric information, like to the V1 cortex in biological vision, which processes basic visual features such as edges and orientations. CSM demonstrates reliable detection of major structural changes, identifying building constructions, infrastructure modifications, and geometric changes across diverse environments. The module shows effectiveness in capturing large-scale geometric discontinuities and edge transitions.

However, the CSM primarily relies on geometric features and lacks sensitivity to directional and texture information, limiting its effectiveness in complex environments such as farmland patterns or intricate road networks. As shown in Fig. 16(2a) and (2b), it fails to achieve precise boundary detection, producing blurred transitions compared to the ground truth Fig. 16(1a). Metrics in Fig. 17 of (C1) further highlight its limited capacity for fine-grained changed areas. To overcome these shortcomings, we design the FODM, enabling multiscale feature detection.

TABLE V
ABLATION STUDY ON MODEL-DRIVEN COMPONENTS

Configuration	Acc	Prec	Rec	F1	mIoU
FODM only	0.771	0.409	0.925	0.567	0.627
CSM only	0.855	0.479	0.763	0.589	0.614
CSM+FODM	**0.906**	**0.685**	**0.821**	**0.747**	**0.719**

2) Effect of the FODM: The FODM is designed to address the CSM's limitations in processing directional and texture information. By incorporating fractional-order differential parameters, this module detects features across multiple scales, drawing inspiration from the coordination mechanisms of the V1 cortices in the HVS.

As illustrated in Fig. 16(3c) and (3d), boundary delineation and texture detection improve compared to the CSM. Metrics in Fig. 17 of (C2) reveal substantial improvements in OA and recall, reflecting reduced unchanged areas detection and enhanced fine-grained changed areas detection. However, the encoding of high-level semantic information remains a challenge, and this challenge is addressed in this article through the design of the cognitive attention module.

In summary, while FODM achieves high recall by capturing rich textural features, it suffers from low precision due to feature confusion in unchanged areas. The introduction of CSM provides essential geometric constraints that establish clear boundaries, effectively suppressing the FPs caused by FODM's texture sensitivity. As evidenced by the density distribution, integrating CSM with FODM reduces high-confidence misclassifications, allowing the model to achieve a synergistic balance between precision and recall, thereby demonstrating their complementary roles within the feature extraction framework, as shown in Table V and Fig. 18.

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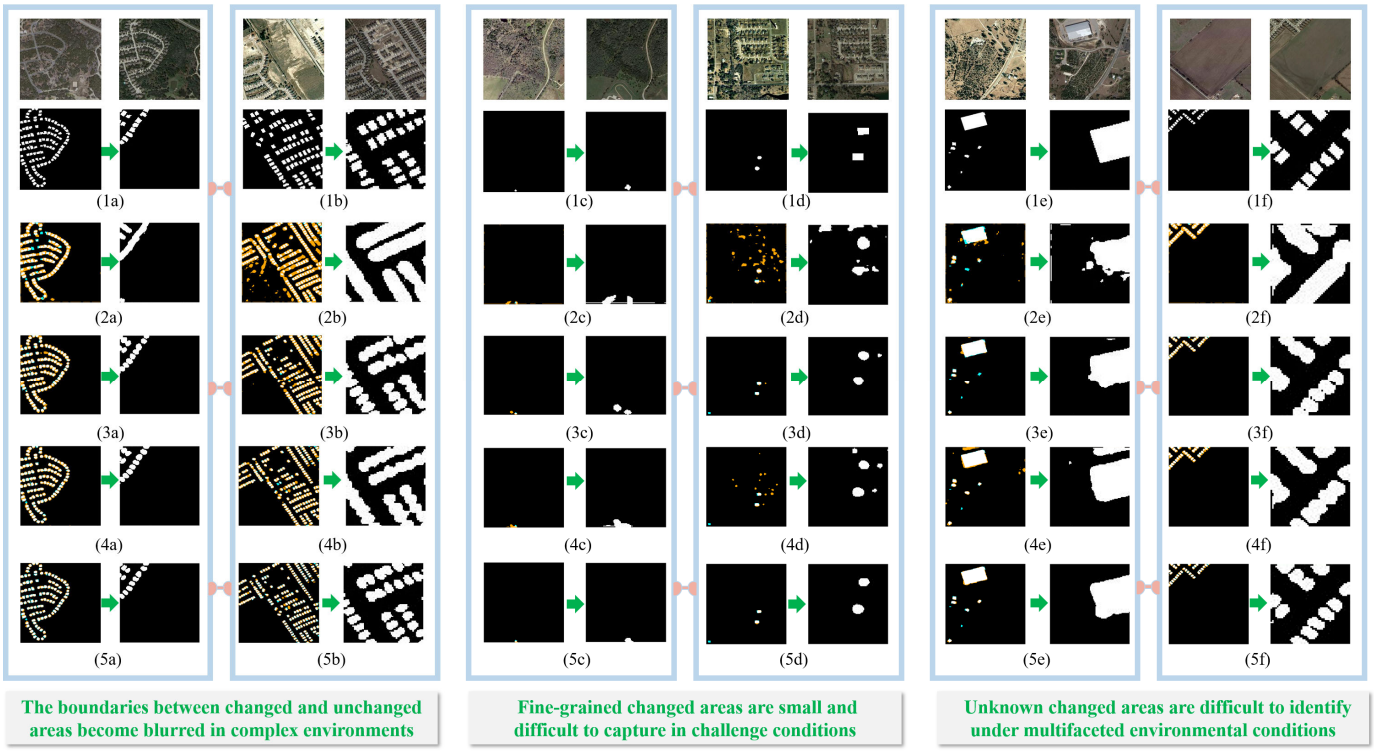


Fig. 16. Results of six image configurations in complex environments. Each result includes (left) a full image and (right) a magnified local area, with green arrows indicating their correspondence. The rows correspond to: 1) Ground truth; 2) CSM; 3) CSM + FODM; 4) CSM + FODM + CAM; and 5) CSM + FODM + CAM + KDH. The columns represent different challenges (a) and (b) blurred boundaries, (c) and (d) fine-grained changes, and (e) and (f) unknown changes under complex environmental conditions [white pixels indicate true positives (TPs), where changed areas are correctly detected; orange pixels represent FPs; and blue pixels denote false negatives (FNs)].

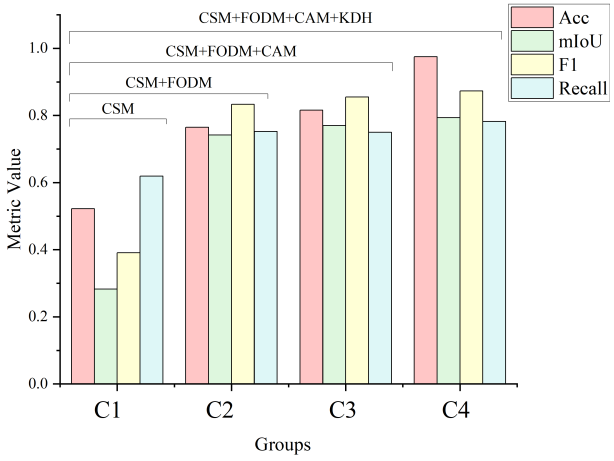


Fig. 17. Performance comparison of modules (CSM, FODM, CAM, and KDH) across different metrics.

3) Effect of the Cognitive Attention Module: The CAM extends the FODM's capabilities by designing mechanisms for high-level semantic feature detection. Inspired by the V4 cortex in the HVS, the CAM enhances semantic features such as color, shape, and motion-induced pseudo-changes through data-driven learning mechanisms.

As shown in Fig. 16(4c) and (4e), it achieves clearer boundaries and fewer FPs compared to the FODM. Metrics in Fig. 17 of (C3) demonstrate consistent improvements across all evaluation criteria. However, integrating the low-level geometric features detected by the CSM and FODM with the high-level semantic features processed by the CAM remains challenging. This problem is addressed by the KDH.

4) Effect of the KDH: The KDH resolves the integration challenges between the model-driven features detected by the CSM and FODM, and the data-driven features processed by the CAM. By employing feedback mechanisms, the KDH refines low-level features while essentially fusing them with high-level semantic information, balancing interpretability and learning capacity.

As shown in Fig. 16(5a)–(5f), it achieves sharper boundaries and more accurate detections compared to the CAM. Metrics in Fig. 17 of (C4) confirm that the KDH outperforms all previous modules, achieving the highest overall performance.

5) Contribution of V1 and V4 Channels in Different Resolution on CDD Data: The CDD dataset contains images with varying resolutions. Since the dataset does not provide explicit per-resolution labels, we design a pixel-dimension-based ground sampling distance (GSD) estimation procedure to assign each image to a resolution group systematically. Based on the physical relationship that, for a fixed ground coverage area A_g, pixel count scales as $N \propto A_g/\mathrm{GSD}^2$, images with finer GSD exhibit larger pixel areas than those with coarser GSD. Accordingly, we applied K-means clustering ($k = 3$) to the pixel area (width $\times$ height in pixels) of each image. The resulting three clusters were ranked in ascending order of centroid value and mapped to resolution groups

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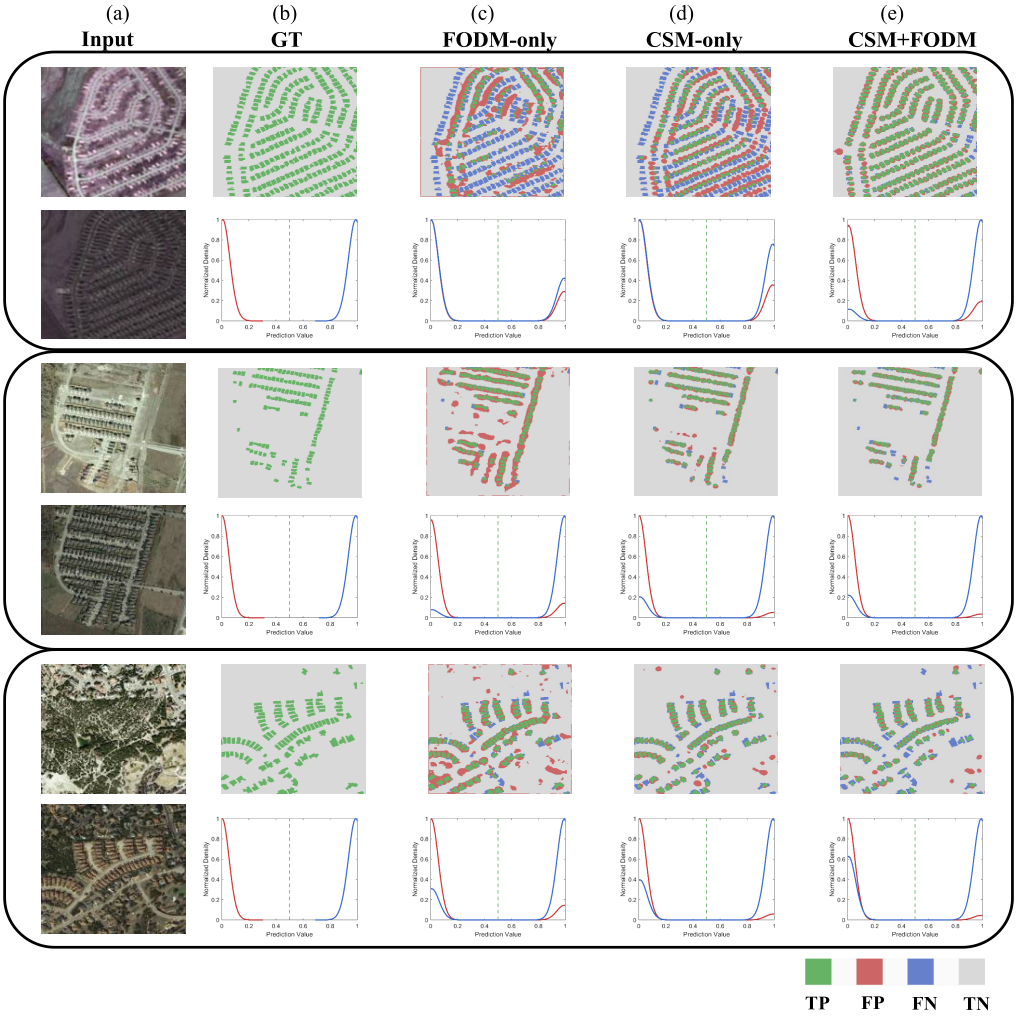


Fig. 18. Qualitative comparison and prediction density analysis of ablation configurations. (a) Input image pairs. (b) GT. (c) FODM-only. (d) CSM-only. (e) CSM + FODM. In the prediction maps, color coding indicates: true positives (TPs, green), false positives (FPs, red), false negatives (FNs, blue), and true negatives (TNs, gray). In the prediction density plots, the red and blue curves represent the normalized density of prediction values for the actual unchanged and changed pixels.

TABLE VI
SAMPLE DISTRIBUTION ACROSS RESOLUTION GROUPS IN THE CDD DATASET

Resolution	Training	Validation	Test	Total
∼0.3 m/pixel	507	161	36	704
∼0.6 m/pixel	568	193	40	801
∼1 m/pixel	625	189	44	858
Total	1,700	543	120	2,363

as follows: the smallest pixel area cluster was assigned to ∼1 m/pixel, the intermediate cluster to ∼0.6 m/pixel, and the largest pixel area cluster to ∼0.3 m/pixel.

The high-resolution images (approximately 0.3 m/pixel) display clear details such as building edges, ground textures, and small changed areas. Medium-resolution images (approximately 0.6 m/pixel) show moderate details with visible road and vehicle outlines but fewer ground texture details. Low-resolution images (approximately 1 m/pixel) exhibit blurring, with boundaries of changed areas being less distinct and texture information being limited. This resolution spectrum allowed us to evaluate our model's performance across varying levels of detail availability. Rather than using the entire CDD dataset, we selectively extracted balanced subsets from each resolution group and partitioned them into training, validation, and test splits using a stratified strategy to preserve the relative resolution distribution across all splits. The complete sample distribution is reported in Table VI.

In Table VII, the 0.6- and 1-m/pixel groups show similar values in three of the four metrics, and the small difference in $F1$-score (94.31% versus 92.47%) can be explained by two main reasons. Primarily, under a coarser GSD, spatial averaging tends to generate smoother and more compact boundaries in the ground truth annotations. This effect is related to the morphology of the annotated change masks and is different from the visual clarity of the input images. Image-level visual clarity decreases as the spatial resolution becomes coarser. Furthermore, at 1-m/pixel resolution, the effective receptive

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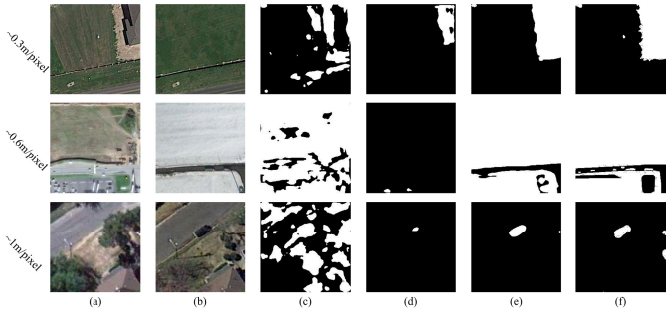


Fig. 19. Different resolution examples in CDD. (a) HR image at time T1. (b) HR image at time T2. (c) Without V1. (d) Without V4. (e) Overall performance. (f) Label.

TABLE VII
PERFORMANCE WITHOUT V1 CHANNEL

Resolution	Metrics			
	OA	F1	mIoU	Recall
∼0.3 m/pixel	45.17 ↓54.4%	61.04	28.02	58.62
∼0.6 m/pixel	50.74 ↓47.6%	50.11	33.62	62.71
∼1 m/pixel	67.19 ↓30.7%	54.61	35.88	74.58

The percentage decreases (in blue) are calculated relative to the OA of the complete model (refer to Table VII) using: (Complete OA − OA without V1)/Complete OA × 100%.

field of the model corresponds to a larger ground area. This property provides slightly stronger contextual information, which is helpful for detecting large-scale change patterns, such as urban expansion and car. The qualitative results in Fig. 19 further support this explanation.

In Table VI, the 0.6-m/pixel group shows near-zero F1 (6.02%) and mIoU (5.83%), while the OA remains at 63.61%. This result indicates a prediction collapse, where the model assigns almost all pixels to the no change class. The high OA is mainly supported by the dominant no-change category, whereas the near-zero F1 demonstrates a failure in detecting changed areas. This phenomenon should provide clear quantitative evidence for the necessity of the CAM module. At 0.6-m/pixel resolution, the spectral variation within change regions becomes more complex, and the model-driven modules cannot effectively handle this variation without the feature alignment introduced by CAM. As a result, removing CAM leads to a complete failure at this resolution.

The importance of CAM is also demonstrated by the qualitative results in Fig. 16. The transition from the CSM + FODM configuration (row 3) to the CSM + FODM + CAM configuration (row 4) produces the obvious improvement among all ablation settings. Changed regions that are entirely missed in row 3 are correctly detected in row 4 after introducing CAM. The further addition of KDH (row 5) brings additional improvement and leads to the final output of the complete model. This result confirms that the proposed framework

TABLE VIII
PERFORMANCE WITHOUT V4 CHANNEL

Resolution	Metrics			
	OA	F1	mIoU	Recall
∼0.3 m/pixel	68.61 ↓30.7%	40.19	37.28	53.33
∼0.6 m/pixel	63.61 ↓34.3%	6.02	5.83	50.19
∼1 m/pixel	87.37 ↓9.8%	52.76	48.61	54.98

TABLE IX
OVERALL PERFORMANCE OF COMPLETE MULTICHANNEL MODEL

Resolution	Metrics			
	OA	F1	mIoU	Recall
∼0.3 m/pixel	98.99	96.98	94.25	97.45
∼0.6 m/pixel	96.78	92.47	86.53	92.11
∼1 m/pixel	96.89	94.31	86.57	92.11

maintains stable and consistent performance across different resolution groups when all modules are integrated.

a) Interpretation of the resolution performance ordering: The approximately 0.3-m/pixel group achieves the highest performance across all four metrics in Tables VII–IX, which is consistent with the common understanding that finer spatial resolution provides richer discriminative features for CD. The approximately 1 m/pixel and approximately 0.6-m/pixel groups produce equivalent results across three of four metrics in Table IX. The small difference in $F1$-score (94.31% versus 92.47%) can be explained by two complementary factors: At coarser GSD, spatial averaging produces geometrically simpler changed region boundaries in the ground truth annotations. This refers specifically to the morphology of annotated change masks, which is a separate property from image-level visual clarity. Furthermore, the model's ground-domain effective receptive field at approximately 1 m/pixel provides slightly better contextual coverage for the large-scale changed areas that are dominant in CDD, as shown in Fig. 19. These findings validate our design inspired by the HVS, confirming that the proposed method enhances overall performance across different resolution levels in the CDD dataset.

To verify the necessity of the proposed KDH module, we compare it with several direct feature combination strategies, including concatenation (Concat), additive fusion (Add), and nonaligned fusion (NoAlign), applied to features from the model-driven modules (CSM and FODM) and the data-driven module (CAM). As reported in Table X, all direct combination methods lead to inferior performance compared with the proposed KDH. Specifically, the Concat strategy incurs higher memory consumption (24.1 ± 0.8 MB) than KDH (11.8 ± 0.3 MB), while achieving lower accuracy (84.8% ± 1% versus 86.1% ± 0.7%). Additive and NoAlign further degrade accuracy, indicating that naive feature merging

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TABLE X
COMPUTATIONAL COMPLEXITY AND ACCURACY ANALYSIS OF MODULES

Method	Cost		Acc.
	Mem. (MB)	Time (ms)	(%)
Module Combinations:			
CSM+FODM+CAM (Concat)	24.1±0.8	8.52±0.15	84.8±1.2
CSM+FODM+CAM (Add)	14.9±0.5	8.60±0.12	83.2±0.9
CSM+FODM+CAM (NoAlign)	13.2±0.6	8.40±0.18	81.6±1.1
Ablations:			
CAM only	19.1±0.4	8.56±0.10	75.4±0.8
CSM+FODM only	34.2±1.2	9.68±0.25	48.5±2.1
Proposed:			
CSM+FODM+CAM+KDH	11.8±0.3	8.53±0.08	86.1±0.7

TABLE XI
ABLATION STUDY OF DIFFERENT INTEGRATION STRATEGIES IN THE CAM MODULE

Strategy	Metrics (%)			
	Accuracy	Precision	Recall	F1
Baseline	96.35	68.35	52.97	59.68
Attention Only	96.02	78.30	30.35	43.75
Simple Attention	95.23	52.37	71.38	60.42
Global Attention	97.25	74.53	69.79	72.09
Channel Attention	92.18	36.40	71.46	48.23
Spatial Attention	97.25	75.09	68.93	71.88
Proposed Method	**97.19**	**81.99**	**73.43**	**77.50**

fails to effectively reconcile heterogeneous representations. The ablation results show that using CAM or CSM + FODM alone leads to substantial performance drops, highlighting the difficulty of bridging the semantic gap between model-driven and data-driven features. By explicitly modeling feature alignment and interaction, the proposed KDH reduces redundancy and improves representational consistency, resulting in superior accuracy with lower computational cost.

In summary, this ablation study demonstrates that the progressive integration of the FODM, CAM, and KDH incrementally addresses the limitations of the baseline CSM. Each module is inspired by aspects of the HVS. This biologically inspired architecture achieves robust performance in HRRS-CD, particularly under complex environments.

6) Further Discussion: To validate the effectiveness of our proposed selective pathway mechanism and the specific integration scheme within the CAM module, we conducted comprehensive ablation experiments comparing seven different strategies for feature integration and attention application. The experimental setup evaluates various approaches, including baseline (original features only), attention-only, simple attention fusion, global attention, channel attention, spatial attention, and our proposed method. The attention-enhanced features F_{att} are generated through a multihead self-attention mechanism that operates on the deep feature representations, enabling the model to dynamically focus on relevant spatial locations and feature channels that are most indicative of changes between bi-temporal images.

The quantitative results presented in Table XI demonstrate performance differences among the evaluated strategies. Our proposed method achieves the highest $F1$-score of 77.50% and precision of 81.99% while maintaining competitive recall of 73.43%, effectively balancing the detection accuracy and reducing FPs. This precision stems from our HVS-inspired selective pathway design that mimics human visual attention mechanisms, allowing the model to focus on truly relevant changed areas while filtering out noise and unchanged areas. As shown in Fig. 20, the channel attention approach exhibits over-detection with excessive FPs, while global and simple attention methods produce more fragmented detection boundaries compared to our proposed approach. The experimental results conclusively demonstrate that our specific integration of global and hierarchical cognitive weights in the CAM outperforms all alternative approaches, validating the necessity of our HVS-inspired selective pathway design for optimal change detection performance.

TABLE XII
COMPARISON OF MODEL PARAMETERS AND COMPUTATIONAL COST [SORTED BY PARAMS (M)]

Algorithm (Year)	Backbone	Params (M)	FLOPs (G)
IDET (2025) [59]	Transformer	**70.89**	96.10
LRNet (2024) [64]	VGG16	<u>48.71</u>	92.23
FTNet (2022) [66]	Transformer	41.03	**202.79**
SNUNet-CD (2022) [58]	NestedUNet	27.07	<u>123.14</u>
AMDANet (2025) [60]	ResNet18	25.83	77.23
LT-MSF (2025) [65]	ResNet18	17.64	5.24
T-C (2022) [61]	Transformer	11.76	5.09
***Ours*(−)**	ResNet18	11.45	26.72
DTCD (2020) [63]	ResNet18	5.32	41.07
BIT (2022) [62]	ResNet18	3.62	18.25

C. Complexity Comparison

The approach presented in this article is the first one to propose the HRRS-CD method systematically mimicking the hierarchical processing of visual cortex areas V1, V2, and V4. A notable advantage of this method over existing ones is its more lightweight model, requiring 11.45-M parameters and 26.72-G FLOPs, representing a significant improvement in the efficiency-performance tradeoff in Table XII. Compared to recent methods, such as IDET (70.89-M parameters, 96.10-G FLOPs) and FTNet (41.03-M parameters, 202.79-G FLOPs), our model delivers better results with substantially lower resource demands. While methods like T-C, BIT, and DTCD perform well in specific scenarios, they fall short in balancing parameter efficiency and computational cost. These underlines highlight the capability of our method to balance computational efficiency and model complexity, providing a resource-efficient and powerful solution for diverse HRRS-CD tasks.

D. Network Visualization

To better understand our model and demonstrate its performance, we used a pair of bitemporal images to generate feature maps at different stages. These feature maps provide a visual representation of how the model focuses on changed areas throughout the processing stages. First, the bitemporal

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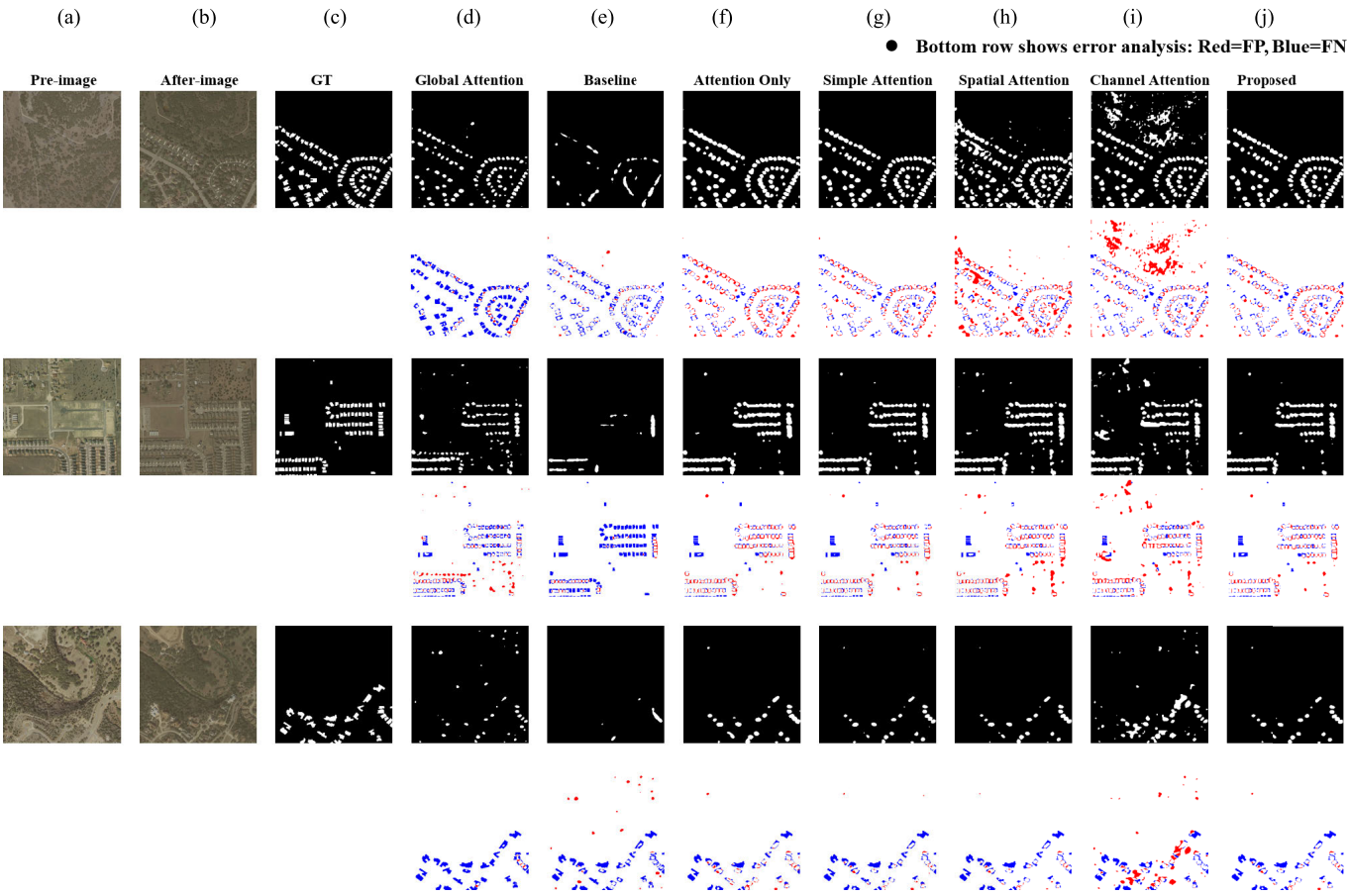


Fig. 20. Visual comparison of different integration strategies. (From left to right) (a) preimages, (b) after-image, (c) GT, (d) global attention, (e) baseline, (f) attention only, (g) simple attention, (h) spatial attention, (i) channel attention, and (j) proposed method.

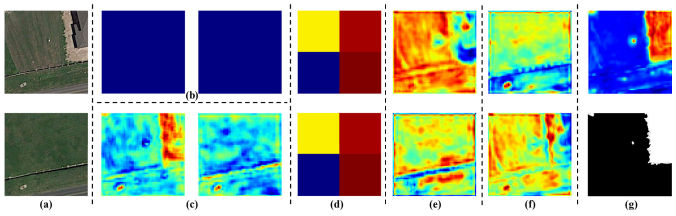


Fig. 21. Network visualization. (a) Input images. (b) Feature maps after the V1 channel. (c) Feature maps after the deep extraction. (d) Token feature maps processed by the V4 channel. (e) Feature maps after fusion. (f) Feature maps after decoder. (g) Final map and ground truth.

images I^t and $I^{t+\Delta t}$ [Fig. 21(a)] are used as input images. The CSM and FODM modules, driven by a model-driven approach, are used to obtain edges and textural feature maps inspired by the V1 channel, denoted as F_{V1} [Fig. 21(b)]. Next, the deep feature extractor captures rich semantic information from the input images at time t and $t + \Delta t$ and generates deep feature maps, denoted as x_i [Fig. 21(c)]. After that, the CAM module, driven by a data-driven approach, integrates dynamic feature extraction and adaptive weight modulation. This process enhances the model's focus on true changed regions while suppressing pseudo-changes, ensuring that the token retains comprehensive and accurate information. As a result, the generated feature maps, inspired by the V4 channel, are denoted as F^i_{V4} [Fig. 21(d)]. To enhance the deep feature maps, we fuse F^i_{V1} and F^i_{V4} to obtain enhanced deep feature maps, denoted as F_{fusion} [Fig. 21(e)]. These feature maps are fed into the decoder function for further prediction, producing x_i^{dec} [Fig. 21(f)]. Finally, the change maps are obtained, denoted as C_{changed} [Fig. 21(g)].

Although all experiments in this work use ResNet18 as the backbone, the proposed modules are designed to work with different backbone networks. The CSM and FODM processes the input images directly before any backbone encoding, so their outputs do not depend on the choice of encoder. The KDH aligns feature maps through a learned regression branch that only requires the feature map dimensions as input, and the loss function supervises the decoder at intermediate scales that are independent of the encoder structure. Therefore, the proposed framework can be integrated with other backbone networks, including transformer-based encoders, without changing the module design. When a stronger backbone such as DINOv2 is used, the richer feature representations are expected to provide better input to the KDH fusion module, while the multiscale edge features extracted by the FODM would remain a useful complement, since such low-level spatial information is not explicitly modeled by foundation-model encoders. Exploring this combination is left as a direction for future work.

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VI. Discussion

This article presents a hierarchical framework inspired by the HVS that effectively addresses feature distribution blurring challenges in complex environments. We analyzed the limitations in existing approaches addressed by our framework. Our method integrates model-driven and data-driven approaches by mimicking the HVS's selective and nonselective pathways. This approach achieves consistent performance improvements and a 3.8% OA gain in complex environments that simultaneously involve seasonal variations, weather conditions, and human activities.

We analyzed which modules are crucial to our results through comprehensive ablation experiments, revealing three key insights. The CSM + FODM combination establishes a geometric processing foundation by simulating the V1 nonselective pathway. When CSM operates independently, OA drops by 30.7%–54.4%, demonstrating its limitation in handling fine-grained changed areas. The FODM module compensates for this by incorporating fractional-order differential operations that preserve directional information. Thus, CSM provides the geometric foundation while FODM adds the directional sensitivity required for changed area edge detection. Furthermore, the CAM module overcomes uniform processing through adaptive semantic discrimination (V4 selective pathway). It focuses on semantically relevant changes while suppressing texture-based false alarms, achieving the highest precision 81.99% among attention strategies. Finally, the KDH integration (V2 pathway) proves most crucial by achieving systematic integration instead of sequential concatenation. It reduces computational overhead from 24.1 $\pm$ 0.8 MB to 11.8 $\pm$ 0.3 MB compared to direct concatenation methods by adaptively weighting features.

Existing methods typically suffer from limitations; model-driven approaches preserve geometric structures but lack semantic adaptability [11], [12], while data-driven methods excel in semantic representation but lose geometric precision [13], [14]. More critically, conventional combination strategies employ sequential concatenation, which fails when pseudo-changes and true changes occupy similar spatial frequencies. Our HVS-inspired framework overcomes these limitations through parallel processing pathways. The nonselective pathway maintains geometric precision, the selective pathway provides semantic discrimination, and the KDH module enables dynamic integration.

In conclusion, our framework currently focuses on binary change detection without providing semantic classification of change types. The hierarchical processing principles developed here could be extended to multiclass change detection by adding more visual cortex mechanisms, such as temporal integration pathways for handling gradual environmental changes [1]. Developing automatic parameter selection for the fractional-order derivatives may improve performance consistency across different environmental conditions. Future work should also investigate attention mechanisms based on visual cortex feedback loops, which could improve the detection of subtle changes in low-contrast environments.

VII. Conclusion

This article presents an HVS-inspired approach for HRRS-CD in complex environments. PCA reveals that seasonal, weather, and anthropogenic variations blur change boundaries, hindering fine-grained detection. We address this with four modules mimicking visual cortex regions: a CSM enhances edges and suppresses noise; a FODM processes multiscale textures; a CAM dynamically weights features; and a KDH integrates model- and data-driven features. Extensive experiments conducted on four benchmark datasets demonstrate that the proposed method achieves consistent and competitive performance while maintaining a compact architecture with 11.45-M parameters. The results suggest that hierarchical integration of physical priors and deep semantic features provides an effective solution for robust CD in complex environments. The HVS-inspired design further offers a generalizable paradigm for combining physical priors with data-driven representations, with potential applicability to other geospatial analysis tasks such as semantic segmentation and object detection in remote sensing images. Nevertheless, the current framework assumes well-registered bi-temporal imagery; significant spatial misalignment affects the effectiveness of the CSM. In addition, the model is restricted to binary change detection and does not explicitly model semantic categories of changes. Future work will focus on incorporating joint image registration and change detection within a unified optimization framework, as well as extending the current binary detection scheme toward semantic change detection to simultaneously identify both the spatial extent and semantic category of changes.

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